

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical

PRACTICAL - 15

AIM : Generating basic summaries using str() or summary() (R).

Input :

R Script: Generating Basic Summaries

Functions: str() and summary()

Dataset: Project Attendance (Simulated)

1. SETUP: Create Sample Project Attendance Data

```
project_df <- data.frame(
```

```
  Student_ID = 1:6,
```

```
  Project_Name = c("AI System", "Web App", "AI System", "Database", "Web App",  
"Database"),
```

```
  Attendance_Percent = c(85.5, 72.0, 91.0, NA, 68.0, 88.0), # Note the NA
```

```
  Submitted = c(TRUE, TRUE, FALSE, TRUE, FALSE, TRUE),
```

```
  Performance_Rating = c(4.5, 3.8, 4.9, 4.0, 3.5, 4.2)
```

```
)
```

```
print("--- Data Loaded ---")
```

```
print(project_df)
```

2. USING str() (Structure)

```
print("--- OUTPUT OF str() ---")
```

```
str(project_df)
```

3. USING summary() (Statistical Summary)

```
print("--- OUTPUT OF summary() [Before Factor Conversion] ---")
```

```
summary(project_df)
```

4. IMPROVING summary() WITH FACTORS

```
project_df$Project_Name <- as.factor(project_df$Project_Name)
```

```
print("--- OUTPUT OF summary() [After Factor Conversion] ---")
```

```
summary(project_df)
```

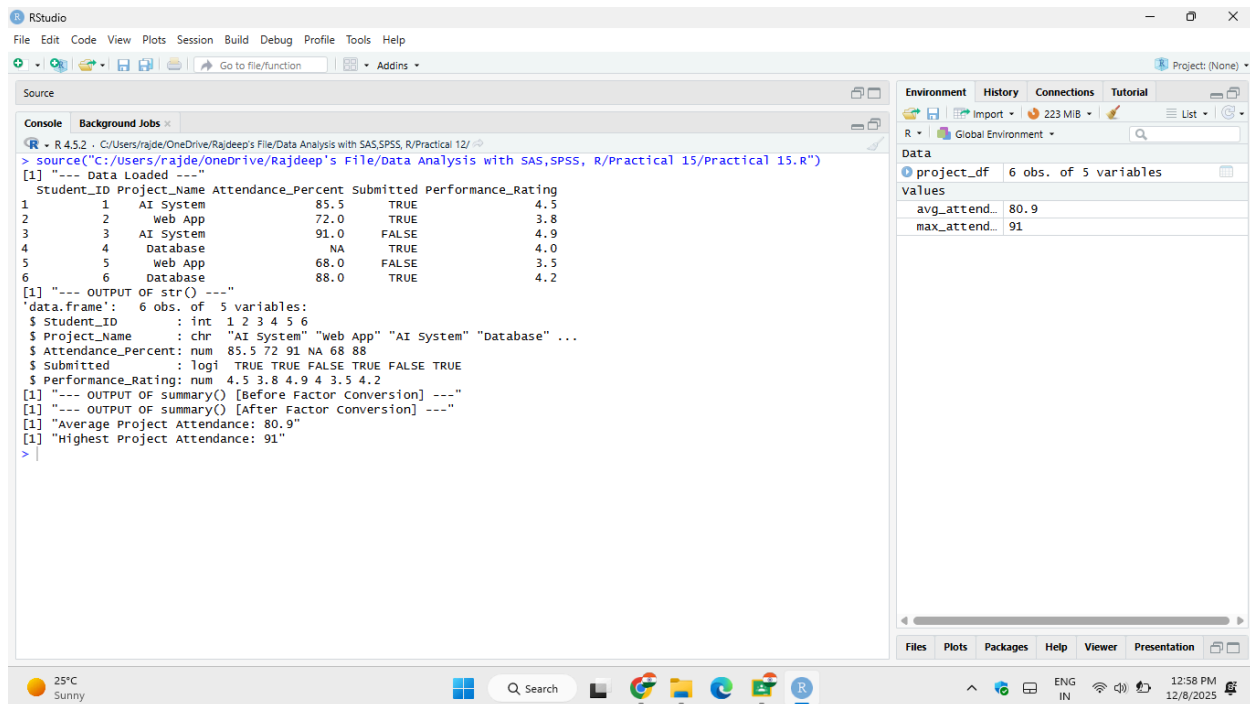
5. Accessing Specific Summaries

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical

```
avg_attendance <- mean(project_df$Attendance_Percent, na.rm = TRUE)
max_attendance <- max(project_df$Attendance_Percent, na.rm = TRUE)
print(paste("Average Project Attendance:", avg_attendance))
print(paste("Highest Project Attendance:", max_attendance))
```

Output :



The screenshot displays the RStudio interface. The console pane on the left shows the execution of the following R code:

```
> source("C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 15/Practical 15.R")
[1] "---- Data Loaded ----"
  Student_ID Project_Name Attendance_Percent Submitted Performance_Rating
1          1    AI System           85.5      TRUE           4.5
2          2    Web App           72.0      TRUE           3.8
3          3    AI System           91.0     FALSE           4.9
4          4   Database            NA      TRUE           4.0
5          5    Web App           68.0     FALSE           3.5
6          6   Database           88.0      TRUE           4.2
[1] "---- OUTPUT OF str() ----"
'data.frame': 6 obs. of 5 variables:
 $ Student_ID      : int  1 2 3 4 5 6
 $ Project_Name    : chr  "AI System" "Web App" "AI System" "Database" ...
 $ Attendance_Percent: num  85.5 72 91 NA 68 88
 $ Submitted       : logi  TRUE TRUE FALSE TRUE FALSE TRUE
 $ Performance_Rating: num  4.5 3.8 4.9 4 3.5 4.2
[1] "---- OUTPUT OF summary() [Before Factor Conversion] ----"
[1] "---- OUTPUT OF summary() [After Factor Conversion] ----"
[1] "Average Project Attendance: 80.9"
[1] "Highest Project Attendance: 91"
>
```

The environment pane on the right shows the data frame 'project_df' with 6 observations and 5 variables. The values for 'avg_attend...' and 'max_attend...' are displayed as 80.9 and 91, respectively.

S101

RAJDEEPP M PARAB

SY.BSc.CS